Introduction to Large Language Models
Assignment- 5

TYPE OF QUESTION: MCQ
Number of questions: 8 **Total mark: 7 X 1 + 1 X 3 = 10**

QUESTION 1: [1 mark]

State True or False.

Lowering the temperature parameter during decoding flattens the output probability distribution and increases the diversity of the generated text.

a. True
b. False

Correct Answer: b

Solution: Lowering the temperature sharpens the distribution and makes generation more deterministic. It is raising the temperature that flattens the distribution and increases diversity.

QUESTION 2: [3 marks]

Given the following encoder and decoder hidden states, compute the attention weights. (Use dot product as the scoring function.)

Encoder hidden states: h1 = [3, 1], h2 = [1, 4], h3 = [2, 2]

Decoder hidden state: s = [1, 1]

a. 0.212, 0.576, 0.212
b. 0.576, 0.212, 0.212
c. 0.333, 0.333, 0.333
d. 0.155, 0.422, 0.423

Correct Answer: a

Solution: e1 = 3(1) + 1(1) = 4
e2 = 1(1) + 4(1) = 5
e3 = 2(1) + 2(1) = 4
alpha1 = e^4 / (e^4 + e^5 + e^4) = 54.598 / 257.609 = 0.212
alpha2 = e^5 / (e^4 + e^5 + e^4) = 148.413 / 257.609 = 0.576
alpha3 = e^4 / (e^4 + e^5 + e^4) = 54.598 / 257.609 = 0.212

QUESTION 3: [1 mark]

What is the primary cause of the vanishing gradient problem in a vanilla RNN?

a. Repeated multiplication by the recurrent weight matrix and by activation derivatives smaller than 1 across many timesteps
b. The absence of any non-linear activation function in the recurrence
c. The use of a fixed-size input window at every timestep
d. The inability of the network to process variable-length sequences

Correct Answer: a

Solution: Backpropagation through time multiplies the same Jacobian at every step. When the relevant terms have magnitude below 1, the product shrinks exponentially with sequence length, so gradients from distant timesteps effectively disappear.

QUESTION 4: [1 mark]

Which gate in an LSTM controls how much of the current cell state is exposed as the hidden state at that timestep?

a. Forget gate
b. Input gate
c. Output gate
d. Cell candidate gate

Correct Answer: c

Solution: The output gate filters the (tanh-squashed) cell state to produce the hidden state h_t. The forget gate decides what to retain from the previous cell state, and the input gate decides how much of the new candidate to write in.

QUESTION 5: [1 mark]

Which limitation of the basic encoder-decoder Seq2Seq model does the attention mechanism directly address?

a. The encoder cannot process variable-length inputs.
b. The whole source sequence has to be compressed into a single fixed-length context vector.
c. The decoder cannot condition on its own previously generated tokens.
d. The model requires the source and target sequences to be of equal length.

Correct Answer: b

Solution: Attention lets the decoder build a fresh, weighted context vector over all encoder hidden states at every decoding step, removing the information bottleneck of a single fixed-length summary vector.

QUESTION 6: [1 mark]

Which of the following is a correct statement about the encoder-decoder architecture?

a. The encoder emits one output token per timestep during decoding.
b. The decoder produces a summary representation of the input sequence.
c. The decoder generates each output conditioned on the encoder representation and on its own previously generated tokens.
d. The encoder retains only the first token of the input sequence.

Correct Answer: c

Solution: Decoding is autoregressive and conditional: at each step the decoder combines information from the encoder with everything it has generated so far.

QUESTION 7: [1 mark]

Why are RNNs preferred over fixed-window feed-forward neural language models?

a. They always have a larger number of parameters.
b. They can process sequences of arbitrary length and share the same weights across timesteps.
c. They remove the need for a word embedding layer.
d. None of the above.

Correct Answer: b

Solution: Please refer to lecture slides.

QUESTION 8: [1 mark]

Which of the following statements about Seq2Seq models are true?

(i) The generation of the target sequence is conditioned on the source sequence.

(ii) In the basic model, the encoder compresses the input into a fixed-size vector representation.

(iii) The input and output sequences must have the same length.

a. (i) and (ii)
b. (ii) only
c. (iii) only
d. (i), (ii) and (iii)

Correct Answer: a

Solution: Seq2Seq models were designed precisely to map an input sequence of one length to an output sequence of a different length, so (iii) is false.
